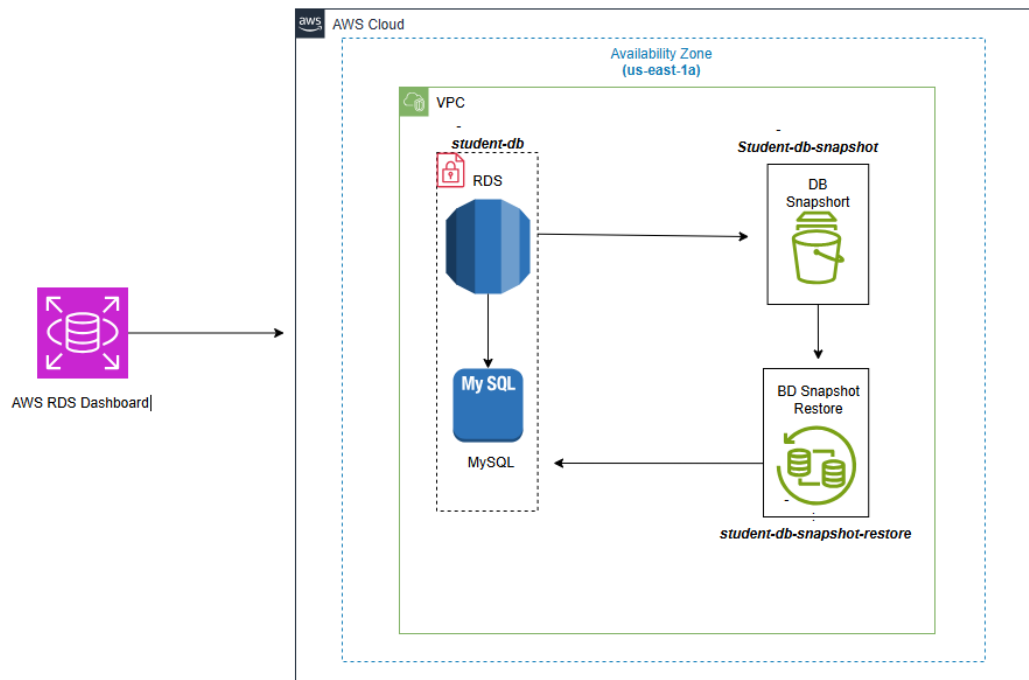


Cloud Databases (Amazon RDS) Labs



Lab 1 - Exploring Amazon RDS

- AWS RDS Dashboard

The screenshot shows the AWS RDS Dashboard. The left sidebar contains navigation links for **Aurora and RDS**, **Dashboard**, **Databases**, **CloudWatch Database Insights**, **Snapshots**, **Exports to Amazon S3**, **Automated backups**, **Reserved instances**, **Proxies**, **Subnet groups**, **Parameter groups**, **Option groups**, **Custom engine versions**, **Zero-ETL integrations**, **Events**, **Event subscriptions**, **Recommendations**, and **Certificate update**.

The main content area includes:

- Welcome to Aurora and RDS**: A section with two cards. The first card, **Create with express configuration in seconds**, allows creating a database with pre-configured settings. The second card, **Create with full configuration**, allows creating a database with custom configurations and enabling features across various engines.
- Service overview**: A section showing metrics for **DB clusters** (0), **DB instances** (1), **Snapshots** (2), and **Recent events** (0). It includes a **View service quota** link.
- Service health**: A section showing the **Current status** of the **Amazon Relational Database Service (N. Virginia)** as **Service is operating normally**. It includes a **View service health dashboard** link.
- Recommended services**: A section showing **No recommendations yet** based on AWS console usage.
- Explore RDS**: A section with a **Start tutorial** link. It includes a **Status** of **Complete** and an **Estimated duration** of **2-5 minutes**.
- RDS costs**: A section showing the **Date range** as **Past 6 months**, the **Region** as **Global**, and the **Credits remaining** as **\$138.6 USD**. It also shows **Days remaining** as **150** (January 7, 2027). A **Cost (\$)** chart is displayed at the bottom.

Lab 2

- Amazon RDS support

Lab 3 - Create Your First Amazon RDB Database

Method

- **Standard Create (Full configurations – no standard creates)**

Engine

- **MySQL** | Template: **Free Tier**

Database Identifier

- **student-db**

Master Username | Password

- **admin** | **1kFGvjx8o6pC**

Database Name:

- **academy**

Public Access

- **No**

Create database [Info](#)

☐ Free plan has access to limited features and resources
The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. [Learn more](#)

Upgrade plan [L](#)

Engine options
Engine type [Info](#)

☒ Aurora (MySQL Compatible)


☐ Aurora (PostgreSQL Compatible)


☒ MySQL


☐ PostgreSQL


☐ MariaDB


☐ Oracle


☐ Microsoft SQL Server


☐ IBM Db2


Choose a database creation method

☒ Full configuration
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Templates
Choose a sample template to meet your use case.

☐ Production
Use defaults for high availability and fast, consistent performance.

☐ Dev/Test
This instance is intended for development use outside of a production environment.

☒ Free tier
Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.

DB instance identifier

Info

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

student-db

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username

Info

Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure

RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed

Create your own password or have RDS create a password that you manage.

☐ Auto generate password

Amazon RDS can generate a password for you, or you can specify your own password.

Master password

Info

Password strength

Very strong

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / " * @

Confirm master password

Info

Public Access

- **No**

Public access

Info

☐ Yes

RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☒ No

RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

Database created

Databases (1)										
Q. Filter by databases										
☐ DB Identifier	☐ Status	☐ Role	☐ Engine	☐ Upgrade rollout ...	☐ Region ...	☐ Size	☐ Recommendations	☐ CPU	☐ Current...	☐ Mainte...
☐ student-db	☒ Backing-up	Instance	MySQL Community	SECOND	us-east-1a	db.t4g.micro		5.44%	0 Conn	none

Lab 4 – Explore the Database Configuration

Audit database instance metadata, networking topology & storage settings

Engine

- **MySQL Community**

Availability Zone

- **us-east-1a**

student-db				
Summary DB identifier student-db CPU 5.52%	Status Available Class db.t4g.micro	Role Instance Current activity 0 Connections	Engine MySQL Community Region & AZ us-east-1a	Recommendations

Storage

- **20 GiB**

VPC

- **vpc-0e9c534171f690a08**

Instance	Instance class	Primary storage	Monitoring
Configuration DB instance ID student-db Engine version 8.4.9 RDS Extended Support Disabled DB name - License model General Public License Option groups default:mysql-8-4  Amazon Resource Name (ARN) arn:aws:rds:us-east-1:208073622332:db:student-db Resource ID db-ph4a6c7n0a7u7ydk2mf5j3tq6aa Created time August 04, 2026, 20:49 (UTC+02:00) DB instance parameter group default:mysql8-4  Deletion protection Disabled Architecture settings Non-multitenant architecture	 Instance class db.t4g.micro vCPU 2 RAM 1 GB Availability Master username admin Master password ***** IAM DB authentication Not enabled Multi-AZ No Secondary Zone -	 Encryption Enabled AWS KMS key aws/rds  Storage type General Purpose SSD (gp2) Storage 20 GiB Provisioned IOPS - Storage throughput - Storage autoscaling Enabled Maximum storage threshold 1000 GiB Storage file system configuration Current	 Monitoring type Database Insights - Standard Database counter metrics Disabled Enhanced Monitoring Disabled

Endpoint

- **student-db.cyf4o0qgmrxxy.us-east-1.rds.amazonaws.com:3306**

Connectivity & security	Networking	Security
Endpoint & port Endpoint student-db.cyf4o0qgmrxxy.us-east-1.rds.amazonaws.com Port 3306	Availability Zone us-east-1a VPC vpc-0e9c534171f690a08 Subnet group default-vpc-0e9c534171f690a08 Subnets subnet-08d027355e39ffb61 subnet-00836247e66e3ae8b subnet-0f13cbb28f16f1ff0b subnet-0a600f4687a8d4977 subnet-08d3db1154d775f9d subnet-096e6f1df863e729c Network type IPv4	VPC security groups default (sg-010a6cbff151c7547)  Publicly accessible No Certificate authority Info rds-ca-rsa2048-g1 Certificate authority date May 26, 2061, 01:34 (UTC+02:00) DB instance certificate expiration date August 04, 2027, 20:48 (UTC+02:00)

Lab 5 – Understanding Backups

Are backups enabled?

- **Yes**

How many days are backups retained?

- **1 Day**

Why are backups important?

- They guarantee business continuity, regulatory compliance & recovery against accidental data corruption or malicious deletion.

student-db ⓘ Ⓐ Modify Actions

Summary		Status	Role	Engine	Recommendations
DB identifier student-db		Available	Instance	MySQL Community	
CPU 4.18%		Class db.t4g.micro	Current activity 0 Connections	Region & AZ us-east-1a	

Connectivity & security | Monitoring | Logs & events | Configuration | Zero-ETL integrations | **Maintenance & backups** | Data migrations | Tags | Recommendations

Maintenance	Maintenance window	Pending maintenance	Pending modifications
Auto minor version upgrade Enabled	August 07, 2026 08:58 - 09:28 (UTC+02:00)	none	
Upgrade rollout order second			

Pending maintenance (0) ⓘ Apply now Apply at next maintenance window

Filter by pending maintenance

Description	Type	Status	Apply date
No pending maintenance available			

Backup	Latest restore time	Replicate to Region
Automated backups Enabled (1 Day)	August 04, 2026, 21:03 (UTC+02:00)	-
Copy tags to snapshots Enabled	Backup window 04:54-05:24 UTC (GMT)	Replicated automated backup -
Backup target AWS Cloud (US East (N. Virginia))		

Delete student-db instance
✕

Permanently delete **student-db** DB instance. You can't undo this action.

Proceeding with this action will delete the instance with all its content and can affect related resources. [Learn more](#)

☒ Create final snapshot
Determines whether a final DB Snapshot is created before the DB instance is deleted.

Final snapshot name
The identifier of the new DB snapshot that is created.

☒ Retain automated backups
Determines whether retaining automated backups for 1 day after deletion

You will be billed for retained backup storage at the rate described as 'Additional backup storage' found in [Backup Storage](#).

To avoid accidental deletion provide additional written consent.

To confirm deletion, type *delete me* into the field.

Cancel
Delete

Lab 6 – Create a Manual Snapshot

Actions → Take Snapshot → Name: student-db-snapshot

Snapshot

- Student-db-snapshot

Take DB Snapshot

Preferences

To take a DB Snapshot, choose a database and name your DB Snapshot.

DB instance

DB Instance Identifier. This is the unique key that identifies a DB Instance.

student-db

Snapshot name

Identifier for the DB Snapshot.

student-db-snapshot

Snapshot identifier is case insensitive, but stored as all lower-case, as in "mysnapshot". Cannot be null, empty, or blank. Must contain from 1 to 255 alphanumeric characters or hyphens. First character must be a letter. Cannot end with a hyphen or contain two consecutive hyphens.

Cancel

Take snapshot

Snapshot created

Successfully created snapshot student-db-snapshot. View details

Snapshots

Manual

System

Shared with me

Public

Backup service

Exports in Amazon S3

Manual snapshots (1)

Filter by manual snapshots

Snapshot name

Engine version

DB instance or cluster

Snapshot creation time

DB Instance created time

Status

Progress

VPC

Snapshot type

Storage

student-db-snapshot

8.4.9

student-db

August 04, 2026, 21:17 (UTC+02:00)

August 04, 2026, 20:49 (UTC+02:00)

Available...

Completed

vpc-0e9c534171f690a08

DB Instance

20 GiB

Snapshot details

student-db-snapshot

Details

ARN

arn:aws:rds:us-east-1:208073622332:snapshot:student-db-snapshot

Instance/Cluster Name

student-db

Master username

admin

DB snapshot name

student-db-snapshot

Snapshot type

manual

DB engine

mysql

DB engine version

8.4.9

License model

general-public-license

Architecture settings

Non-multitenant architecture

Option group

default:mysql-8-4

Zone

us-east-1a

KMS key ID

a3c1ed9c-496e-452e-9def-a808b3773d70

Source region

N/A

Snapshot Creation Time

August 04, 2026, 21:17 (UTC+02:00)

Original Snapshot Creation Time

August 04, 2026, 21:17 (UTC+02:00)

Instance/Cluster Creation

August 04, 2026, 20:49 (UTC+02:00)

Backup retention period

1 day

Backup window

04:54-05:24

VPC

vpc-0e9c534171f690a08

Status

Available

Storage type

General Purpose SSD (gp2)

DB storage

20 GiB

IOPS

-

Storage throughput

0

Port

3306

Time zone

Dedicated Log Volume

Turned off

Lab 7 Restore from a Snapshot

Select Snapshot → Choose: Restore Snapshot

Restore snapshot

You are creating a new DB instance or DB cluster from a snapshot. The default VPC security group and parameter group are selected for the new DB instance or DB cluster, but you can change these settings.

Free plan has access to limited features and resources

The free plan limits the features and resources that are available for RDS and Aurora databases. Upgrade your account plan to remove all limitations. [Learn more](#)

[Upgrade plan](#)

DB instance settings

DB engine

MySQL Community

☐ Enable RDS Extended Support [Info](#)

Amazon RDS Extended Support is a [paid offering](#). By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the [RDS for MySQL documentation](#).

Availability and durability

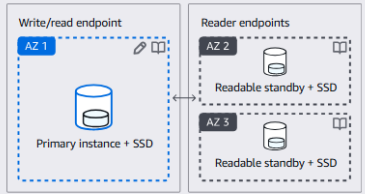
Deployment options [Info](#)

Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the [Amazon RDS service level agreement \(SLA\)](#).

Multi-AZ DB cluster deployment (3 instances)

Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:

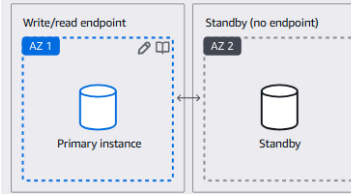
- 99.99% uptime
- Redundancy across Availability Zones
- Increased read capacity
- Reduced write latency



Multi-AZ DB instance deployment (2 instances)

Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:

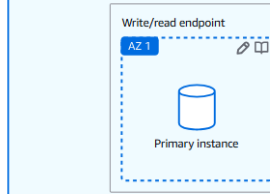
- 99.99% uptime
- Redundancy across Availability Zones



Single-AZ DB instance deployment (1 instance)

Creates a single DB instance without standby instances. This setup provides:

- 99.9% uptime
- No data redundancy



Restore Snapshot

- DB Identifier: ***student-db-snapshot-restore***

Snapshot being restored

Snapshot is being restored Restoring student-db-snapshot to student-db-snapshot-restore. This might take several minutes.									
Databases (1)									
Filter by databases									
DB Identifier	Status	Role	Engine	Upgrade rollout ...	Region ...	Size	Recommendations	CPU	Current...
student-db-snapshot-restore	Creating	Instance	MySQL Community	SECOND	-	db.t3.micro		-	none

Snapshot restored

Snapshot restored successfully Restoring student-db-snapshot to student-db-snapshot-restore was successful.									
Databases (1)									
Filter by databases									
DB Identifier	Status	Role	Engine	Upgrade rollout ...	Region ...	Size	Recommendations	CPU	Current...
student-db-snapshot-restore	Available	Instance	MySQL Community	SECOND	us-east-1f	db.t3.micro		8.36%	0 Conn

Lab 8 – Monitoring Amazon RDS

CPU Utilization

- **7.22%**

Summary				
DB Identifier student-db-snapshot-restore	Status Available	Role Instance	Engine MySQL Community	Recommendations
CPU 7.22%	Class db.t3.micro	Current activity 0 Connections	Region & AZ us-east-1f	

Database Connection

- **student-db-snapshot-restore.cyf4o0qgmrxy.us-east-1.rds.amazonaws.com:3306**

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations

Tags

Recommendations

Connect using

Code snippets

Use when connecting through SDK, APIs, or third-party tools including agents.

CloudShell

Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

Endpoints

Use when connecting through any IDE interface.

Database name

mysql

Master username

admin

Internet access gateway

Disabled

Port

3306

Endpoint type

Instance endpoint

Additional configurations

Connectivity & security

Endpoint & port

Endpoint

student-db-snapshot-restore.cyf4o0qgmrxy.us-east-1.rds.amazonaws.com

Port

3306

Networking

Availability Zone

us-east-1f

VPC

vpc-0e9c534171f690a08

Subnet group

default-vpc-0e9c534171f690a08

Subnets

subnet-08d027555e39ff061

subnet-00836247e66e3ae8b

subnet-0f13c028f16f1ff0b

subnet-0a0204607a04a8f77

subnet-08d5d81154d775f9d

subnet-096e6f1df863e729c

Network type

IPv4

Security

VPC security groups

default (sg-010a6c0f151c7547)

Active

Publicly accessible

No

Certificate authority

rds-ca-rs2048-g1

Certificate authority date

May 26, 2061, 01:34 (UTC+02:00)

DB instance certificate expiration date

August 10, 2027, 22:42 (UTC+02:00)

Free Storage Space

Primary storage

Encryption

Enabled

AWS KMS key

aws/rds

Storage type

General Purpose SSD (gp2)

Storage

20 GiB

Provisioned IOPS

-

Storage throughput

-

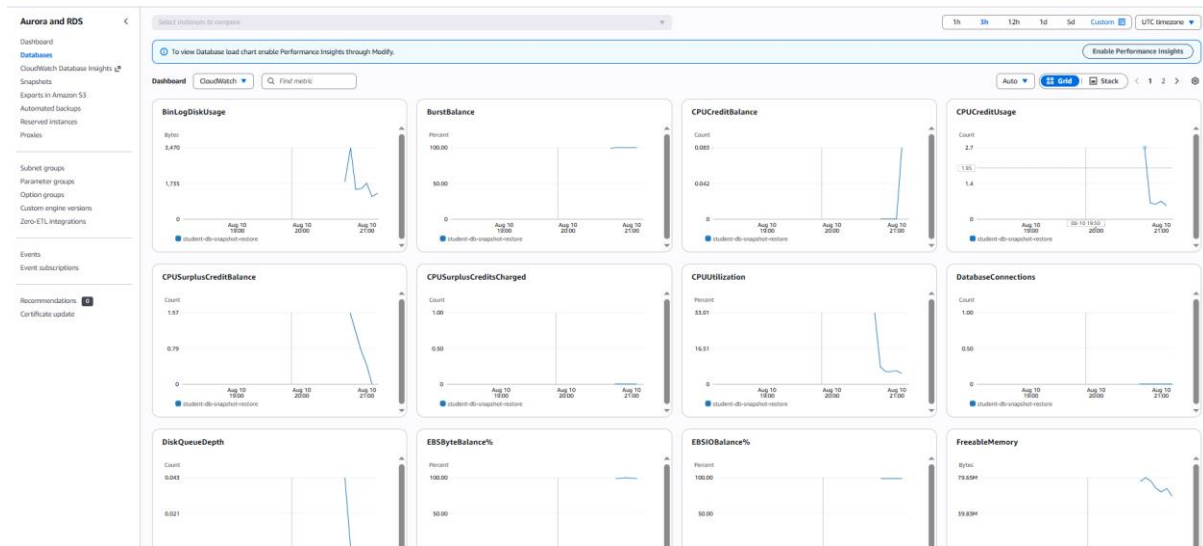
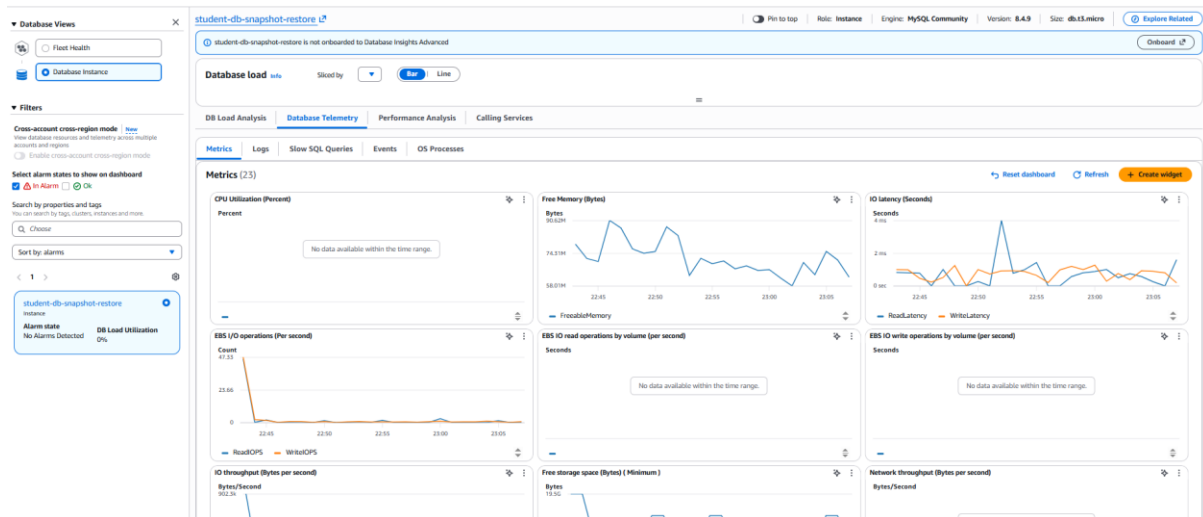
Storage autoscaling

Disabled

Storage file system configuration

Current

Read/Write Latency



Lab 9 - Explore Amazon RDS Pricing

RDS costs



Date range: Past 6 months

Region: Global

Costs in your free plan account are covered by credits.

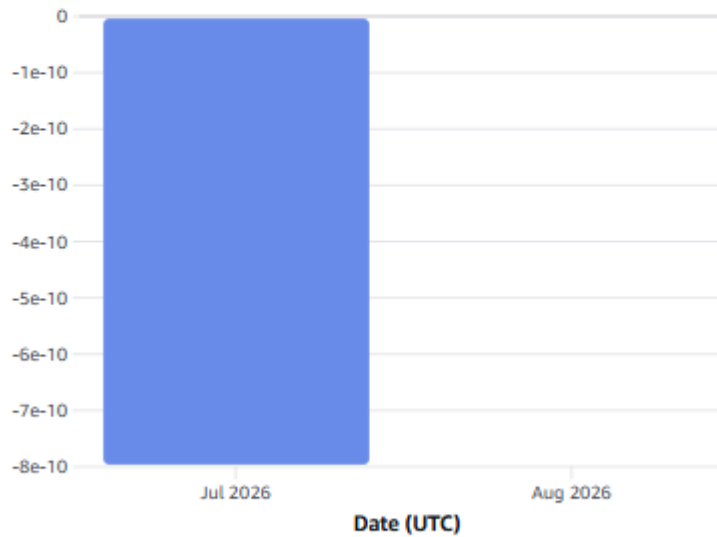
Credits remaining

\$138.56 USD

Days remaining

150 (January 7, 2027)

Cost (\$)



■ us-east-1

[Analyze your costs in Cost Explorer](#)

Lab 10